MEDINOVA TEACHING HOSPITAL

Class 10 — Endocrinology

DKA · HHS · Thyroid Emergencies · Adrenal Crisis

> *"Hormones govern silently until they fail catastrophically."*

SENIOR RESIDENCY SERIES | MODULE 10 of 20

SECTION 01

Diabetic Ketoacidosis (DKA) Protocol

Diagnosis: Glucose > 11 mmol/L (or known DM), Ketones > 3.0 mmol/L, and pH < 7.3 (or Bicarbonate < 15 mmol/L).

Fluid Replacement Schedule (Example 70kg adult)

Fluid Number	Volume (0.9% Saline)	Duration
1st Liter	1000 ml	Over 1 hour
2nd & 3rd Liter	1000 ml each	Over 2 hours each
4th & 5th Liter	1000 ml each	Over 4 hours each
6th Liter	1000 ml	Over 6 hours

Potassium Replacement Table

Note: Do not start insulin if K+ < 3.3 mmol/L.

Serum K+ (mmol/L)	Potassium Addition to 1L bag
> 5.5	None
3.5 – 5.5	40 mmol/L
< 3.5	Senior Review - may need central line / intensive care

Insulin Infusion: Fixed rate at 0.1 units/kg/hr. Continue long-acting (basal) insulin. Switch to 5% Dextrose alongside Saline when Glucose < 14 mmol/L to prevent hypoglycaemia while clearing ketones.

> **Prof. Obi's Pearl #1** "You are treating the acidosis, not the glucose. Do not stop the insulin infusion until the pH is > 7.3 and the ketones are < 0.6, even if the sugar is normal."

SECTION 02

DKA vs. HHS Comparison

HHS (Hyperosmolar Hyperglycaemic State) carries a higher mortality than DKA due to extreme dehydration and older age at presentation.

Feature	DKA (Type 1 DM)	HHS (Type 2 DM)
Onset	Rapid (< 24 hours)	Gradual (Days/Weeks)
Glucose	> 11.0 mmol/L	> 30.0 mmol/L
Osmolality	Variable	> 320 mOsm/kg
Ketones	Elevated (++ / > 3.0)	Low / Absent
Fluid Deficit	3 - 6 Liters	8 - 12 Liters
Management	Fixed-rate insulin	Fluid resuscitation first; lower insulin rate.

SECTION 03

Hypoglycaemia Management

Whipple's Triad: 1. Symptoms of hypo, 2. Low plasma glucose (< 3.9 mmol/L), 3. Resolution of symptoms with sugar.

1. Mild/Moderate (Conscious): 15-20g quick-acting carbs (e.g., 4-5 GlucoTabs).
2. Severe (Unconscious): 200ml 10% Dextrose IV bolus OR 1mg Glucagon IM.
3. Post-Recovery: Long-acting carb (e.g., slice of toast) to prevent recurrence.

> Prof. Obi's Pearl #2 "The most common cause of recurrent hypoglycaemia in a hospital is the failure to give a complex carbohydrate after the initial rescue. A bolus of dextrose only buys you 20 minutes."

SECTION 04

Thyroid Emergencies

1. Thyroid Storm (Hyperthyroid)

Use the **Burch-Wartofsky Point Scale** (Temp, CNS, GI-Hepatic, CVS, Heart Failure, History). Score > 45 is highly suggestive.

Intervention	Drug & Action
Step 1: Block Action	Propranolol (reduces HR and peripheral T4-T3 conversion).
Step 2: Block Synthesis	Propylthiouracil (PTU) or Carbimazole.
Step 3: Block Release	Lugol's Iodine (must give 1h *after* PTU).
Step 4: Block Conversion	Hydrocortisone.

2. Myxoedema Coma (Hypothyroid)

Presents as Hypothermia, Bradycardia, and altered mental state. Rx: IV Levothyroxine + IV Hydrocortisone (to cover potential co-existing adrenal insufficiency) + Gentle warming.

SECTION 05

Adrenal Crisis

Identify in patients with known Addison's or long-term steroid use presenting with hypotension, vomiting, and abdominal pain.

Acute Rx: 100mg Hydrocortisone IV stat → 200mg/24h infusion. Aggressive IV Saline resuscitation.
Sick Day Rules: Double oral steroid dose during illness/fever. If vomiting → IM Hydrocortisone and seek ED.

SECTION 06

Diabetes Management & Medications

Class	Mechanism	Key Benefit / Risk
Metformin	↑ Insulin sensitivity	Weight neutral. GI side effects.
SGLT2i (Gliflozins)	Glycosuria (urine)	CV/Renal protection. Risk of eDKA/UTIs.
GLP-1 RA (Tides)	Incretin mimic	Significant weight loss. Nausea.
Sulphonylureas	↑ Insulin secretion	Effective but high Hypo risk.

> **Prof. Obi's Pearl #3** "If a patient on an SGLT2 inhibitor is unwell, check their ketones even if their glucose is normal. Euglycaemic DKA is a silent killer."

SECTION 07

Secondary Endocrine Hypertension

1. Conn's Syndrome (Primary Aldosteronism):
• Triad: Hypertension, Hypokalaemia, Metabolic Alkalosis.
• Screen: Aldosterone-to-Renin Ratio (ARR).

2. Phaeochromocytoma:
• Symptoms: Triad of episodic headache, sweating, and palpitations.
• Screen: 24h Urinary Metanephrines or Plasma Metanephrines.

3. Cushing's Syndrome:
• Features: Moon face, central obesity, purple striae, proximal myopathy.
• Screen: Overnight Dexamethasone Suppression Test or 24h Urinary Free Cortisol.

> **Prof. Obi's Pearl #4** "Endocrinology is the medicine of feedback loops. Never interpret a hormone result without looking at its stimulating or inhibiting counterpart."

Medical Definitions:
• **eDKA:** Euglycaemic DKA (Acidosis with normal sugar).
• **Metanephrines:** Breakdown products of adrenaline; used to detect adrenal tumors.
• **Sick Day Rules:** Patient instructions for managing chronic conditions during acute illness.
• **Basal-Bolus:** Insulin regimen mimicking physiological secretion.

OFFICIAL SIGN-OFF

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SERIES: Senior Residency Module 10 of 20
DATE: March 2026